

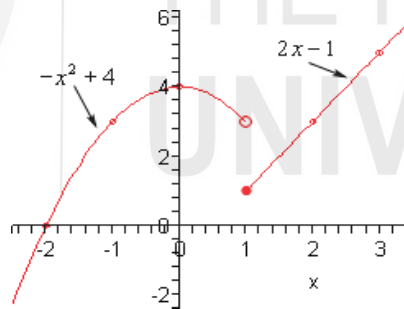
$$-x^2 \text{ on } x < 1 \text{ and}$$

$$2x-1 \text{ on } x \geq 1$$

Use  $x = 1$  in both functions not overlooking to the point at  $x = 1$  since technically that will only be valid for the bottom function. The table of values for each function on the specified range is:

| $x$ | $f(x) = -x^2 + 4$ | $(x, y)$  |
|-----|-------------------|-----------|
| -2  | 0                 | $(-2, 0)$ |
| -1  | 3                 | $(-1, 3)$ |
| 0   | 4                 | $(0, 4)$  |
| 1   | 3                 | $(1, 3)$  |
| $x$ | $2x-1$            | $(x, y)$  |
| 1   | 1                 | $(1, 1)$  |
| 2   | 3                 | $(2, 3)$  |
| 3   | 5                 | $(3, 5)$  |

Sketch the graph. For the top function we used an open dot for the point at  $x = 1$  and for the bottom function we used a closed dot at  $x = 1$ . In this way we make it clear on the graph that only the bottom function really has a point at  $x=1$ .



Note that two graphs did not meet at  $x = 1$  and there a blank space in the graph.

**Q6.** Sketch the graph of the following piece wise function.

$$x + 3 \text{ if } x < 2$$

$$h(x) = \begin{cases} x^2 & \text{if } -2 \leq x < 1 \\ -x+2 & \text{if } x \geq 1 \end{cases}$$

**Ans.** In this case we will graph three functions on the ranges given above. So, as with the previous example we will get function values for each function in its specified range and we will include the endpoints of each range in each

computation. When we graph we will acknowledge which function the end point actually belongs with by using a closed dot as we did previously. Also, the top and bottom functions are lines and so we do not need more than two points for these two. We will get a couple more points for the middle function.

| <b>X</b> | <b>x+3</b>           | <b>(x, y)</b> |
|----------|----------------------|---------------|
| -3       | 0                    | (-3, 0)       |
| -2       | 1                    | (-2, 1)       |
| <b>x</b> | <b>x<sup>2</sup></b> | <b>(x, y)</b> |
| -2       | 4                    | (-2, 4)       |
| -1       | 1                    | (-1, 1)       |
| 0        | 0                    | (0, 0)        |
| 1        | 1                    | (1, 1)        |
| <b>x</b> | <b>-x + 2</b>        | <b>(x, y)</b> |
| 1        | 1                    | (1, 1)        |
| 2        | 0                    | (2, 0)        |

**Q7.** Given the following equation  $9x^2 - 16y^2 = 144$

- Find the  $x$  and  $y$  intercepts, if possible, of the graph of the equation.
- Find the coordinates of the foci.
- Sketch the graph of the equation.

**Ans:** a)  $9x^2/144 - 16y^2/144 = 1$  or  $x^2/16 - y^2/9 = 1$  or  $x^2/4^2 - y^2/3^2 = 1$

The equation is that of a hyperbola with  $a=4$  and  $b=3$ .

To get the  $x$  intercept, set  $y=0$  in the equation. Thus,  $x^2/4^2 = 1$  and  $x = \pm 4$ .

For they intercept set  $x=0$ . So  $-y^2/3^2 = 1$ . Without any real solution of the equation, we do not get  $y$  intercept.

- Find  $c$  first.

$Asc^2 = a^2 + b^2$ , from (a) values of  $a$  and  $b$  can be put to obtain

$$c^2 = 4^2 + 3^2 \text{ or, } = 25$$

Thus, foci are  $F_1(5, 0)$  and  $F_2(-5, 0)$ .

- 

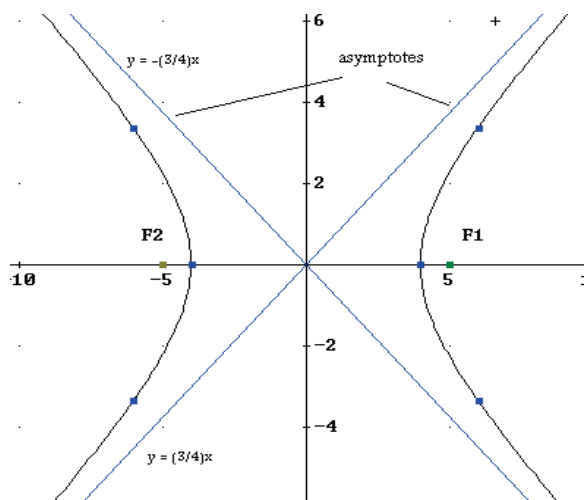
- Find the asymptotes  $y = -(b/a)x$  and  $y = (b/a)x$  and plot them.

Thus,  $y = -(3/4)x$  and  $y = (3/4)x$

Then plot the  $x$  intercepts

- set  $x=6$  and find  $y$  such that  $9(6)^2 - 16y^2 = 144$  or,  $-16y^2 = 144 - 324$  or,  $y^2 = 45/4$  or,  $y = 3(5)^{1/2}/2$  and  $y = -3(5)^{1/2}/2$ .

Graph of hyperbola is:



**Q8.** Suppose you have following rational function:  $y= \frac{2x^2 -18}{x^2-4}$ . Draw its graph.

**Ans.** Before you draw a T-chart, first find the asymptotes and intercepts. Then choose  $x$ 's between the  $x$ -intercepts and vertical asymptotes, to give you the additional information necessary to graph the function.

The vertical asymptotes are at  $x = 2$  and  $x = -2$ , the horizontal asymptote is at  $y = 2$ , the  $x$ -intercepts are at  $x = -3$  and  $x = 3$ , and the  $y$ -intercept is at  $y=4.5$ . Now, plot a few additional points between these other points:

| $x$  | $y = \frac{2x^2 - 18}{x^2 - 4}$ |
|------|---------------------------------|
| -5   | 1.5                             |
| -2.5 | -2.4                            |
| -1   | 5.3                             |
| 1    | 5.3                             |
| 2.5  | -2.4                            |
| 5    | 1.5                             |

